

Wenyu Chiou

Artificial Intelligence Evaluation & Behavioral Simulation Research Engineer

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Research engineer evaluating large language model (LLM) agents against measured human behavior and building quantitative behavioral simulations. Anchored by a **937-household** survey and a **52,141-household** coupled model.

Open to Summer 2027 internships in AI research, LLM evaluation, agent development, and computational social science.

Work authorization: F-1 student; CPT-eligible for internships.

— EXPERIENCE

Graduate Research Assistant — Complex Adaptive Water Systems Lab, Lehigh University Aug 2024 – present

- Designed a human-versus-LLM study using **937** household responses, multi-group structural equation modeling (SEM), and label-blind socioeconomic personas to compare generated behavior with measured pathways; planned submission to *Progress in Disaster Science*.
- Designed and fielded a **937-household** flood-adaptation survey; used SEM to identify social-group pathway differences and Bayesian calibration to translate observed pathways into agent parameters.
- Built **FLOODABM**, a **52,141-household** coupled agent-based × catastrophe simulation across 27 census tracts and 2011–2023; validated outcomes against OpenFEMA claims over 50 Monte Carlo runs.
- Designed and substantially completed **WAGF**, a six-stage pipeline checking LLM-agent decisions against domain rules and behavioral theory before state updates; research software in preparation for release.

Independent Research Engineering & Open Source 2024 – present

- Built an **evaluation suite** for research-hub’s literature search that checks retrieval recall against expected papers; retained observed failures as explicit audit baselines instead of deleting them.
- Shipped **research-hub**, a local-first tool with Model Context Protocol (MCP) and command-line interfaces (CLIs); published **research-hub-pipeline v1.1.1** on PyPI, tested cross-platform, and contributed **18 merged pull requests**.
- Created and self-hosted awesome-agentic-ai-zh, a trilingual agentic-AI roadmap with **240+ resources**, **23 exercises**, and **77 MCP/Skill entries**; reached **5K+ stars** and **700+ forks**.

— SELECTED RESEARCH OUTPUTS

- Yang, Y. C. E., & **Chiou, W.** (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. DOI [Published](#)
- Chiou, W.**, et al. Household Flood Adaptation and Financial Outcomes: A Coupled Human–Flood Modeling Analysis of Homeowners and Renters. *Water Resources Research*. *Under revision following peer review*
- Chiou, W.**, et al. Flood Adaptation Decision Variables and Pathways across Social Groups: A Comparison between Human Survey Responses and Responses Generated by Large Language Models. Planned submission to *Progress in Disaster Science*.

— SELECTED CONFERENCE PRESENTATIONS

- International Society for Data Science and Analytics (ISDSA) 2026 Annual Meeting** — “What Makes AI Believable?” Comparing AI and human responses in mental-health risk assessment; recorded presentation.
- American Geophysical Union (AGU) Fall Meeting 2025** — Poster NH41E-0449: “Modeling Long-Term Household Flood Adaptation under Social Heterogeneity: A Coupled Agent-Based Modeling Framework.”
- The 2nd International Sociohydrology Conference (ISHC) 2025**, Tokyo — Oral Presentation 38-03: “Investigating the Decision-Making Process and Causal Relationships between Flood Risk Perception and Adaptation Actions.”

— EDUCATION

- Ph.D., Civil & Environmental Engineering**, Lehigh University 2024 – present · expected 2028
- M.S., Hydrological and Oceanic Sciences**, National Central University Sep 2021 – Jun 2023
- B.S., Earth Sciences**, National Central University Sep 2017 – Jun 2021

— METHODS & TECHNICAL SKILLS

LLM / agent systems: LLM application & agent development · AI-system evaluation · evaluation infrastructure · prompt/context engineering · retrieval-augmented generation (RAG) · memory · tool use · multi-agent workflows · MCP

Methods: computational social science · behavioral simulation · experimental design · agent-based modeling (ABM) · structural equation modeling (SEM) · Bayesian calibration

Tools: Python · R · pytest / continuous integration (CI) · Git · OpenAI / Anthropic APIs · Ollama · Windows / POSIX